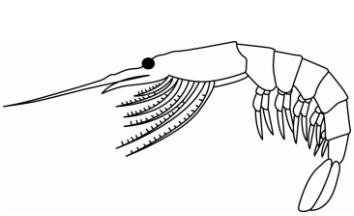
Short questions

|!|BCESC0290000001|!|

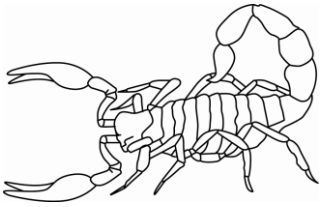
The diagrams below show six kinds of arthropods.



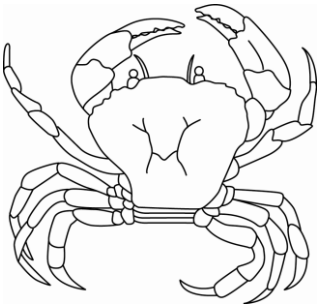
A



B



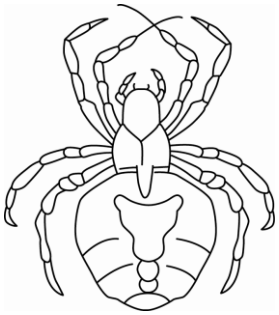
C



D



E



F

(a) State ***one*** feature common to arthropods A to F. (1 mark)

_____

_____

(b) B and E belong to the same major arthropod group. State ***two*** features that distinguish this arthropod group. (2 marks)

_____

_____

_____

_____

(c) The dichotomous key below can be used for identifying arthropods A to F by their genera.

1a	three pairs of legs	go to 2
1b	four or more pairs of legs	go to 3
2a	one pair of wings	*Musca*
2b	two pairs of wings	*Anax*
3a	pincers at the front end of body	go to 4

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3b	no pincers at front end of body	go to 5
4a	tail held above body	*Buthus*
4b	tail tucked under body, or not visible	*Cancer*
5a	four pairs of legs	*Aranea*
5b	more than four pairs of legs	*Pandalina*

Using the key, identify which genera arthropods B, D and F belong to. Write the key sequence taken for making the identification. (3 marks)

Arthropod	Genus name	Key sequence for identification
B	(a)	(b)
D	(c)	(d)
F	(e)	(f)

##

(a) They do not have a backbone / vertebral column / have an exoskeleton / have jointed legs / antennas. (1)

(b) Their bodies are divided into three main parts (the head, thorax and abdomen). (1)
They have three pairs of legs. (1)

(c) 1 mark for each row correct (3)

Arthropod	Genus name	Key sequence for identification
B	(a) *Musca*	(b) 1a → 2a
D	(c) *Cancer*	(d) 1b → 3a → 4b
F	(e) *Aranea*	(f) 1b → 3b → 5a

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Analysing data	3

|!|BCESB0290000002|!| ★

The photographs below show two different plants, A and B.

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A



[**Photo source:** Goldi64 / cropped / CC-BY-SA-3.0]

B



[**Photo source:** Biswarup Gangulyb / CC-BY-SA-3.0]

(a) Which of plants A and B is a monocotyledonous plant and which is a dicotyledonous plant? Explain your answers. (4 marks)

(b) Plants A and B belong to the same major plant group. Give ***three*** common features of the plants in this group. (3 marks)

##

(a) Plant A is a monocotyledonous plant. (1)
Its leaves have parallel veins. / The number of petals in the flower is in a multiple of three. (1)
Plant B is a dicotyledonous plant. (1)
Its leaves have a branching network of veins. / The number of petals in the flower is five. (1)

(b) Any ***three*** of the following: (3)

- They produce flowers for reproduction.

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- They produce seeds protected inside fruits.
- They have true root, stems and leaves.
- They have vascular tissues.

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Understanding basic concepts	3

|!|BCESB0290000003|!| ★ **Cross topic**

Animals can be classified into vertebrates and invertebrates. Vertebrates have a backbone while invertebrates do not.

(a) Give ***one*** function of the vertebral column. (1 mark)

(b) Use the dichotomous key below to identify the five groups of vertebrates. (5 marks)

1a	with fins and gills	Group A
1b	without fins and gills	go to 2
2a	lay eggs with hard shells	go to 3
2b	do not lay eggs with hard shells	go to 4
3a	with feathers	Group B
3b	without feathers	Group C
4a	with hair	Group D
4b	without hair	Group E

Group A: _____

Group B: _____

Group C: _____

Group D: _____

Group E: _____

##

(a) Any ***one*** of the following: (1)

- encloses and protects the spinal cord
- helps supports the body weight

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- provides attachments for girdles and muscles for movement

(b) Group A: fish (1)
Group B: birds (1)
Group C: reptiles (1)
Group D: mammals (1)
Group E: amphibians (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	5

|!|BCESC0290000004|!|

The table below shows some features of four organisms belonging to different kingdoms.

Organism	Cell wall made of cellulose	Cell wall made of peptidoglycan	Chlorophyll	Mitochondria
P	✓	×	✓	✓
Q	×	×	×	✓
R	×	✓	✓	×
S	×	×	✓	×

Key: ✓ = present × = absent

(a) Match P, Q, R and S with the respective kingdoms to which they may belong. Put the letters A to D in the space provided. (4 marks)

Organism		***Kingdom***
P	_____	A. Eubacteria
Q	_____	B. Archaebacteria
R	_____	C. Fungi
S	_____	D. Plantae

(b) Give ***one other*** feature that can be used to distinguish between R and S. (1 mark)

_____

_____

_____

_____

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##

(a) P: D (1)

Q: C (1)

R: A (1)

S: B (1)

(b) The lipids in their cell membranes (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Understanding basic concepts	1

|!|BCESB0290000005|!| ★

The photographs below show three different animals.



X



Y



Z

(a) Give a reason for classifying the three animals in the same phylum. (1 mark)

(b) Two of the animals shown belong to the same group while the other belongs to a different group.

(i) Which two animals belong to the same group? (1 mark)

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(ii) State ***two*** structural features which show that the two animals in (i) belongs to this group. (2 marks)

(iii) What is the difference between the reproduction of these two groups of animals? (1 mark)

##

(a) They have a backbone / vertebral column. (1)

(b) (i) Y and Z (1)

(ii) Any ***two*** of the following: (2)

- They have hair covering their bodies.
- They have mammary glands.
- They have pinnae.

(iii) X lays eggs with hard shells while Y and Z give birth to live young. (1)

(5 marks)

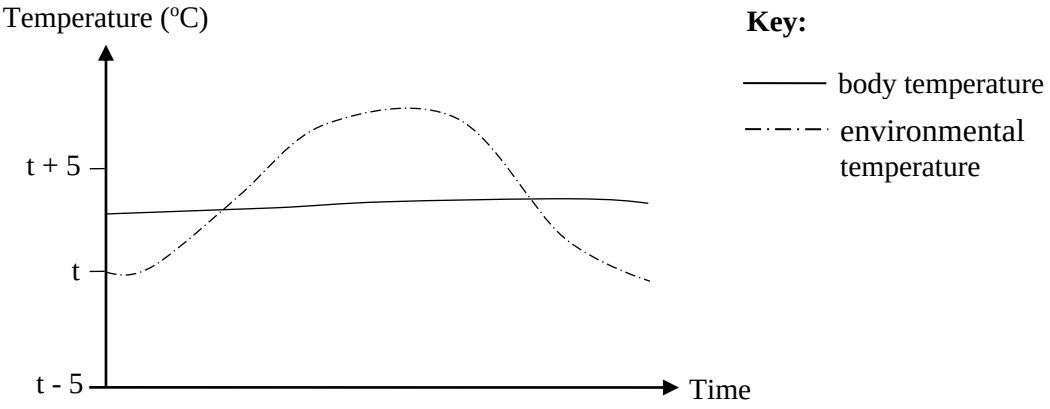
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	1

|!|BCESB0290000006|!|

The graph below shows the changes in the body temperature of a vertebrate, X, and the environmental temperature over a period of time.

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(a) Determine whether X is an ectotherm or endotherm. Explain your answer. (2 marks)

(b) Name ***two*** possible vertebrate groups to which X may belong. (2 marks)

(c) (i) State ***one*** common feature, other than body temperature regulation, of the two vertebrate groups in (b). (1 mark)

(ii) State ***two*** features that allow us to distinguish the two vertebrate groups in (b). (2 marks)

##

(a) X is an endotherm (1)
because its body temperature remains relatively constant despite variations of environmental temperature. (1)

(b) mammals, birds (1, 1)

(c) (i) Any ***one*** of the following: (1)
- Both groups carry out internal fertilization.
- Both groups are heterotrophs.

(Accept other reasonable answers.)

(ii) Any ***two*** of the following: (2)
- Birds have feathers covering their bodies while mammals have hair.

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- The embryo of birds develops inside a shelled egg while the embryo of most mammals develop inside the mother's body.
- Mammals have mammary glands while birds do not.

(Accept other reasonable answers.)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2

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